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# nginx\_passwd

*Release 3.3.2*

**Gene C**

**May 05, 2026**



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## NGINX-PASSWD

### 1.1 Overview

Basic Auth Password Manager.

Manages basic auth password files. This replaces the functionality provided by `htpasswd` from Apache. It also provides legacy and modern hash functions, including the `argon2` and `pbkdf2_sha512`.

- All git tags are signed with [arch@sapience.com](mailto:arch@sapience.com) key which is available via WKD or download from <https://www.sapience.com/tech>. Add the key to your package builder gpg keyring. The key is included in the Arch package and the `source=` line with *?signed* at the end can be used to verify the git tag. You can also manually verify the signature

### 1.2 New Or Interesting

#### Version 3.3.0

- Source code reorg
- Switch python packager from hatch to uv
- Confirm all working with python 3.14.2

#### Version 3.1.0

- Remove `os.getlogin()` in tests (ENOTTY reported by @ccharabaruk on AUR package).
- Small tidy ups

#### Version 3.0.0

- Reduce our reliance on `passlib` (it is unmaintained).
  - `passlib bcrypt` is broken
  - there is a fork which works
- Algo changes - limit use of `passlib` only to legacy hashes
  - `bcrypt` : now uses `python-bcrypt` directly
  - `argon2` : now uses `python-argon2-cffi` (which calls C library)
  - `pbkdf2_sha512` : now uses `python-cryptography`
  - `pbkdf2_sha256` : now uses `python-cryptography`
  - `md5_crypt` : legacy algo using `passlib`
  - `apr_md5_crypt` : legacy algo using `passlib`

**Older**

- PEP-8, PEP-257, PEP-484 and PEP 561
- Refactor code
- Add pytest
- add python-bcrypt dependency to Arch PKBUILD
- For simplicity, make argon2 required instead of optional

## GETTING STARTED

### 2.1 nginx\_passwd application

#### 2.1.1 Usage

To add or modify a user and write the resulting password file:

```
nginx-passwd -f <password_file> <user>
```

If file is not specified, then the result is written to stdout.

The supported algorithms are:

```
* Modern : argon2, pbkdf2_sha512, pbkdf2_sha256
* Active  : sha512, sha256, bcrypt
* Deprecated: md5, apr_md5/apr1
```

with *sha256* being the default.

Note: At some point in future we will change the default algo to the more modern *argon2*.

Older and now deprecated algorithms (*md5*, *apr\_md5*) are still supported but should be replaced by one of the active ones.

Note: *apr\_md5*, also known as *apr1*, is the ancient Apache variant of md5.

Note: If the password is not provided with *-p* option and is nowhere is found to read it from, then it will be an empty string.

#### 2.1.2 Options

The options are given below and *nginx-passwd -h* provides a help summary.

Positional Argument:

- username  
required argument.
- *(-h, -help)*  
show help message and exit
- *(-f, -passwd\_file) <password\_file>*  
Write to this Password file

- *(-a, -algo)* <algorithm>

Default is now argon2 (older versions used sha256). Can be one of:

```
* Modern : argon2, pbkdf2_sha512, pbkdf2_sha256
* Active  : sha512, sha256, bcrypt
* Deprecated: md5, md5_apr1
```

- *(-p, -passwd)* <password>

Password as an option. Without this it will be read from stdin.

- *(-D, -delete)*

Delete this user from the password file.

- *(-v, -verify)*

Checks that the provided password matches that in the password file



## 3.1 Installation

Available on \* [Github](#) \* [Archlinux AUR](#)

On Arch you can build using the PKGBUILD provided in packaging directory or from the AUR package. To build manually, clone the repo and do:

Listing 1: Manual Install

```
rm -f dist/*
/usr/bin/python -m build --wheel --no-isolation
root_dest="/"
./scripts/do-install $root_dest
```

When running as non-root then set root\_dest a user writable directory

## 3.2 Dependencies

- Run Time :
  - python (3.13 or later)
  - passlib
  - bcrypt (aka python-bcrypt)
  - argon2-cffi (aka python-argon2-cffi)
  - cryptography (aka python-cryptography)

*NB* versions 1.1 and earlier used openssl - all newer version now use python passlib library.

- Building Package:
  - git
  - hatch (aka python-hatch)
  - wheel (aka python-wheel)
  - build (aka python-build)
  - installer (aka python-installer)
  - rsync
- Optional for building docs:

- sphinx
- texlive-latexextra (archlinux packaguing of texlive tools)

## 3.3 Philosophy

We follow the *live at head commit* philosophy as recommended by Google's Abseil team<sup>1</sup>. This means we recommend using the latest commit on git master branch.

## 3.4 License

Created by Gene C. It is licensed under the terms of the GPL-2.0-or-later license.

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<sup>1</sup> <https://abseil.io/about/philosophy#upgrade-support>

**LICENSE**

nginx\_passwd provides a Basic Auth Password File Manager supporting new and legacy hash algorithms.

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